Primal : $\max \quad 4x_1 - x_2 + 3x_3$

$$\text{s.t.} \quad \begin{aligned} 5x_1 - x_2 + 2x_3 &\le 4 \\ -2x_1 + x_2 + x_3 &\le 3 \\ x_1,\ x_2,\ x_3 &\ge 0 \end{aligned}$$

Dual : $\min \quad 4\pi_1 + 3\pi_2$

$$\text{s.t.} \quad \begin{aligned} 5\pi_1 - 2\pi_2 &\ge 4 \\ -\pi_1 + \pi_2 &\ge -1 \\ 2\pi_1 + \pi_2 &\ge 3 \\ \pi_1,\ \pi_2 &\ge 0 \end{aligned}$$

$\pi = (\frac{4}{3}, \frac{1}{3})$ check :

$$\begin{cases} 5\pi_1 - 2\pi_2 = 6 \ge 4 \\ -\pi_1 + \pi_2 = -1 \ge -1 \\ 2\pi_1 + \pi_2 = 3 \ge 3 \\ \pi_1 = \frac{4}{3} \ge 0 \\ \pi_2 = \frac{1}{3} \ge 0 \end{cases}$$

Therefore, $\pi = (\frac{4}{3}, \frac{1}{3})$ is a feasible solution to the dual

To check whether $\pi = (\frac{4}{3}, \frac{1}{3})$ is the optimal solution

$$\begin{cases} \pi_1 (a_1^T x - b_1) = \frac{4}{3}(5x_1 - x_2 + 2x_3 - 4) = 0 \\ \pi_2 (a_2^T x - b_2) = \frac{1}{3}(-2x_1 + x_2 + x_3 - 3) = 0 \\ (C_1 - \pi^T A_1) x_1 = (4 - 5 \times \frac{4}{3} + 2 \times \frac{2}{3}) x_1 = 0 \\ (C_2 - \pi^T A_2) x_2 = (-1 + \frac{4}{3} - \frac{1}{3}) x_2 = 0 \\ (C_3 - \pi^T A_3) x_3 = (3 - 2 \times \frac{4}{3} - \frac{1}{3}) x_3 = 0 \end{cases} \Longrightarrow x = (0, \frac{2}{3}, \frac{7}{3})$$

check :

$$\begin{cases} 5x_1 - x_2 + 2x_3 = 4 \le 4 \\ -2x_1 + x_2 + x_3 = 3 \le 3 \\ x_1 = 0 \ge 0 \\ x_2 = \frac{2}{3} \ge 0 \\ x_3 = \frac{7}{3} \ge 0 \end{cases}$$

Therefore, $x = (0, \frac{2}{3}, \frac{7}{3})$ is a feasible solution to the primal

All these considered:

Both $x = (0, \frac{2}{3}, \frac{7}{3})$ and $\pi = (\frac{4}{3}, \frac{1}{3})$ are optimal solutions to the primal and dual problems, respectively. $Z = 4x_1 - x_2 + 3x_3 = 4\pi_1 + 3\pi_2 = \frac{19}{3}$